Mathematics

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Overview Programs Courses

CODE	NAME	DESCRIPTION
MATH100A	Mathematics for Elementary Teaching, I	This class teaches ways to think about and explain elementary school mathematics. Topics include: cultural and base-n number systems, algorithms, elementary number theory, probability, and graphing.
MATH100B	Mathematics for Elementary Teaching, II	Completes the explanation of elementary school mathematics by discussing geometry and algebra. Discusses the pedagogy with the California mathematics framework, the NCTM standards, and "replacement units".
MATH101A	Classical Number Systems	A development of classical number systems starting with natural numbers and progressing through the integral, rational, real and complex number systems. Includes an introduction to number theory, mathematical induction, the fundamental theorem of ari...
MATH101B	Mathematical Systems	The theory of operations within rings and fields. Polynomials, the fundamental theorem of algebra and other factorization theorems. Analyzing linear, polynomial, rational and exponential functions. Solving algebraic equations and inequalities. Especi...
MATH102A	Modern Euclidean and Noneuclidean	Topics in plane and solid geometry. The axioms of pure, euclidean, projective, and noneuclidean

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	Geometry	geometry. Transformational geometry (isometries, dilitations, involutions, perspectivities, and projectivities). The history and the historical implicatio...
MATH102B	Modern Euclidean and Noneuclidean Geometry	Topics in plane and solid geometry. The axioms of pure, euclidean, projective, and noneuclidean geometry. Transformational geometry (isometries, dilitations, involutions, perspectivities, and projectivities). The history and the historical implicatio...
MATH103	Introduction to Group Theory	Permutation groups, cyclic groups, theory of finite groups, group homomorphisms and isomorphisms, and Abelian groups. Applications to number theory and geometry.
MATH104A	Introduction Into Numerical Analysis	Numerical methods for the solution of nonlinear equations (Newton method), for integration (quadrature formulas and composite integration), and for the initial value problem for ordinary differential equations (Euler and Kutta methods).
MATH104B	Numerical Analysis	Numerical methods for the solution of systems of linear equations (direct and iteractive methods), and the finite difference methods for boundary value problems for (ordinary and partial) differential equations.
MATH104C	Advanced Topics in Numerical Analysis	Topics in approximation theory; numerical methods for finding eigenvalues of a matrix; and advanced topics in numerical methods for ordinary and partial differential equations.
MATH108A	Linear Algebra	Abstract vector spaces and subspaces. Span and linear independence. Basis and dimension. Linear maps. Eigenvalues and eigenvectors.
MATH108B	Linear Algebra	Diagonalization, inner product spaces, projections, least-squares approximations, invariant factors and elementary divisors, canonical forms, topics from advanced matrix theory, applied linear algebra, and group representation theory.

CODE	NAME	DESCRIPTION
MATH108C	Matrix Analysis	Eigenvalues, eigenvectors and similarity; Unitary equivalence and normal matrices; QR factorization; Least squares; Singular Value decomposition; Moore-Penrose generalized inverse; Hermitian and symmetric matrices; Variational characterization of eig...
MATH111A	Abstract Algebra	An introduction to group theory, focusing on finite groups. Basic algebraic concepts including homomorphisms, normal subgroups, quotients, direct products and group actions. Classification of finitely generated Abelian groups. Other topics such as se...
MATH111B	Abstract Algebra	Ring theory, ideals, factorization. Examples include number rings and polynomial rings. Module theory if time permits.
MATH111C	Abstract Algebra	Fields, field extensions, Galois theory.
MATH113	Non-euclidean Geometry	An introduction to hyperbolic geometry with some discussion of other non-euclidean systems.
MATH114	Computational Tools for Numerical Analysis	Introduction to programming with specific examples in scientific computing. The algorithms used will be analyzed in detail in Math 104A-B-C. The emphasis will be the translation of numerical algorithms into actual working code.
MATH115A	Introduction to Number Theory	Number-theoretic functions, divisibility, congruences, primitive roots and indices, quadratic residues and the quadratic reciprocity law.
MATH115B	Introduction to Number Theory	Selected topics at the discretion of the instructor. Possibilities include (but are not limited to) Diophantine equations, the distribution of primes, number-theoretic methods in cryptography, quadratic forms, continued fractions, algebraic number t...
MATH115C	Topics in Number Theory	Selected topics in number theory at the direction of the instructor.
MATH116	Combinatorial Analysis	Elementary counting principles, binomial coefficients, generating functions, recurrence

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		relations, the principle of inclusion and exclusion, distributions and partitions, systems of distinct representatives, applications to computation.
MATH117	Methods of Analysis	Introduction to methods of proof in analysis. topics include limits, sequences and series, continuity, compactness, as well as other topics. This course is intended to follow Mathematics 8 and to introduce students to the level of sophistication of u...
MATH118A	Real Analysis	Axioms of the real numbers, topology of metric spaces, numerical sequences, infinite series.
MATH118B	Real Analysis	Real functions: continuity, differentiability, Riemann integrals, power and trigonometric series.
MATH118C	Real Analysis	Multivariable calculus and a selection of other topics including differential forms or Lebesgue integral.
MATH119A	Ordinary Differential Equations	Existence, uniqueness, and stability; the geometry of phase space; linear systems and hyperbolicity; maps and diffeomorphisms.
MATH119B	Chaotic Dynamics and Bifurcation Theory	Hyperbolic structure and chaos; center manifolds; bifurcation theory; and the Feigenbaum and Ruelle-Takens cascades to strange attractors.
MATH122A	Theory of Complex Variables	Complex numbers, complex differentiation, analytic functions, Cauchy-Riemann equations, Taylor and Laurent series, classification of singular points, residues, evaluation of improper integrals.
MATH122B	Theory of Complex Variables	The argument principle, Rouche's theorem, analytic continuation, conformal mapping, fractional transformations and applications, infinite products, harmonic functions. Other topics at the discretion of the instructor.
MATH124A	Partial Differential Equations	Wave, heat, and potential equations.
MATH124B	Fourier Series and Numerical Methods	Fourier series; generalized functions; and numerical methods.
MATH132A	Optimization	Global and local optimization, constrained and unconstrained optimization, convex versus

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		nonconvex optimization, Projection Theorem, linear search methods, gradient methods, Newton and Quasi-Newton methods, trust region methods.
MATH132B	Operations Research	Large scale optimization, stochastic gradient descent, inexact Newton methods, limited memory BSGF, least squares problems, global optimization, optimization in general normed spaces.
MATH137A	Graph and Network Theory	Elements of graph and network theory including paths, circuits, trees, coloring, planarity, matching theory, Hall's theorem, applications to scheduling theory, flows in networks, Menger's theorem, and other topics as time permits.
MATH137B	Spectral Graph Theory	The graph Laplacian, eigenvalues of weighted graphs and random walks, the normalized Laplacian and Cheegers inequality, diameters and eigenvalues, expander graphs, eigenvalues of symmetric graphs, heat kernels.
MATH145	Introduction to Topology	Metric spaces, continuity, compactness, classification of surfaces, Euler characteristics, and fundamental groups. Additional topics at the discretion of the instructor.
MATH147A	Differential Geometry of Curves and Surfaces	Curves and surfaces in three-dimensional Euclidean space, including Frenet formulas, signed curvature of plane curves, first and second fundamental forms, normal curvature and principal curvature, Gaussian and mean curvature, Gauss equation and Theor...
MATH147B	Differential Geometry of Curves and Surfaces	Parallel transport, geodesics, Gauss-Bonnet Theorem, fundamental theorem of surfaces. Additional topics at the discretion of the instructor may include minimal surfaces, Codazzi equation, surfaces of constant Gaussian curvature, isothermal coordinate...
MATH170	Introduction to Mathematical Finance	Describes mathematical methods for estimating and evaluating asset pricing models, equilibrium and derivative pricing, options, bonds, and the term-structure of interest rates. Also introduces finance optimization models for risk management and finan...

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MATH181A	Advanced Problem Solving in Mathematical, Historical and Pedagogical Contexts.	Focuses on the representations, strategies, and language learners use to conceptualize and develop fundamental ideas of mathematics. Includes advanced mathematical problem solving and its implications for teaching and learning at the secondary level....
MATH181B	Advanced Problem Solving in Mathematical, Historical, and Pedagogical Contexts	Continuation of Math 181A or ED 134. Focuses on the representations, strategies, and language learners use to conceptualize and develop fundamental ideas of mathematics. Includes advanced mathematical problem solving and its implications for teaching...
MATH182	History of Mathematics	An examination of the major achievements in mathematical thinking throughout history. Topics may include the history of numerical systems in early civilizations, the development of formal proof, mathematical contributions from diverse populations and...
MATH190	Special Topics in Mathematics	An advanced treatment of mathematics content appropriate for undergraduates. The material for the course may come from algebra, geometry, analysis, applied mathematics or other areas of mathematics. Topics vary by instructor.
MATH190AI	Mathematical Reasonings of AI such as GPT and DeepSeek	This project-based course explores the mathematical reasonings of AI systems like GPT and DeepSeek. Students will work on exercises drawn from topics in Math 8, linear algebra, and calculus. For each project, the student will obtain an AI-generated s...
MATH190PS	Problem Solving	The Problem Solving in Mathematics course is designed to impart students with advanced problem-solving techniques and strategies in mathematics. Students will explore a diverse range of methods, including the application of parity, coloring, ordering...
MATH191	Mathematics Transition for Transfer Students	Provides transfer students with a solid mathematical foundation to ease their transition into a four-year college environment. Students will explore the fundamental axioms of integers and their properties,

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		gaining a deep understanding of this essenti...
MATH193	Internship in Mathematics	Faculty sponsored academic internship in industrial or research firms.
MATH194GS	Group Studies for Advanced Students	Participants will select a math-related book or papers, read the section before the next meeting and discuss reading at the meeting. Readings may include biographies of mathematicians, histories or popularizations of mathematics, textbooks, and readi...
MATH197A	Senior Thesis	Independent research under the supervision of a faculty member which will result in a senior thesis. Student will concentrate on reading and gathering material for a thesis.
MATH197B	Senior Thesis	Independent research under the supervision of a faculty member which will result in a senior thesis. Student will concentrate on writing a thesis.
MATH199	Independent Studies in Mathematics	Coursework consists of academic research supervised by a faculty member on a topic not available in established course offerings.
MATH199RA	Independent Research Assistance	Coursework consists of faculty supervised research assistance.
MATH201A	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH201B	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH201C	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH202A	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex

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		analysis.
MATH202B	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex analysis.
MATH202C	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex analysis.
MATH206A	Matrix Analysis and Computation	Graduate level-matrix theory with introduction to matrix computations. SVD's, pseudoinverses, variational characterization of eigenvalues, perturbation theory, direct and iterative methods for matrix computations.
MATH206B	Numerical Simulation	Linear multistep methods and Runge-Kutta methods for ordinary differential equations: stability, order and convergence. Stiffness. Differential algebraic equations. Numerical solution of boundary value problems.
MATH206C	Numerical Solution of Partial Differential Equations--Finite Difference Methods	Finite difference methods for hyperbolic, parabolic and elliptic PDE's, with application to problems in science and engineering. Convergence, consistency, order and stability of finite difference methods. Dissipation and dispersion. Finite volume met...
MATH206D	Numerical Solution of Partial Differential Equations - Finite Element Methods	Weighted residual and finite element methods for the solution of hyperbolic, parabolic and elliptic partial differential equations, with application to problems in science and engineering. Error estimates. Standard and discontinuous Galerkin methods.
MATH214A	Ordinary Differential Equations	Existence, uniqueness, and stability; the geometry of phase space; linear systems and hyperbolicity; maps and diffeomorphisms.
MATH214B	Chaotic Dynamics and Bifurcation Theory	Hyberbolic structure and chaos; center manifolds; bifurcation theory; and the Feigenbaum and Ruelle-Takens cascades to strange attractors.

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MATH215A	Partial Differential Equations	Wave, heat, and potential equations.
MATH215B	Fourier Series and Numerical Methods	Fourier series; generalized functions; and numerical methods.
MATH220A	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH220B	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH220C	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH221A	Foundations of Topology	Metric spaces, topological spaces, continuity, Hausdorff condition, compactness, connectedness, product spaces, quotient spaces. Other topics as time allows.
MATH221B	Homotopy Theory	Homotopy groups, exact sequences, fiber spaces, covering spaces, van Kampen Theorem.
MATH221C	Differential Topology	Topological manifolds, differentiable manifolds, transversality, tangent bundles, Borsuk-Ulam theorem, orientation and intersection number, Lefschetz fixed point theorem, vector fields.
MATH225A	Topics in Number Theory	Selected topics in number theory.
MATH225B	Topics in Number Theory	Selected topics in number theory.
MATH225C	Topics in Number Theory	Selected topics in number theory.
MATH227A	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.

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MATH227B	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.
MATH227C	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.
MATH228A	Functional Analysis	Topics in functional analysis such as operators on Hilbert space, convex analysis, fixed point theorems, distribution theory, unbounded operators.
MATH228B	Functional Analysis	Topics in functional analysis such as operators on Hilbert space, convex analysis, fixed point theorems, distribution theory, unbounded operators.
MATH231A	Lie Groups and Lie Algebras	Differentiable manifolds, definition and examples of lie groups, lie group-lie algebra correspondence, nilpotent and solvable lie algebras, classification of semi-simple lie algebras over the complexes, representations of lie groups and lie algebras,...
MATH231B	Lie Groups and Lie Algebras	Differentiable manifolds, definition and examples of lie groups, lie group-lie algebra correspondence, nilpotent and solvable lie algebras, classification of semi-simple lie algebras over the complexes, representations of lie groups and lie algebras,...
MATH232A	Algebraic Topology	Singular homology and cohomology, exact sequences, Hurewicz theorem, Poincare duality.
MATH232B	Algebraic Topology	Singular homology and cohomology, exact sequences, Hurewicz theorem, Poincare duality.
MATH236A	Homological Algebra	Algebraic construction of homology and cohomology theories, aimed at applications to topology, geometry, groups and rings. Special emphasis on hom and tensor functors; projective, injective and flat modules; exact sequences; chain complexes; derived...
MATH236B	Homological Algebra	Algebraic construction of homology and cohomology theories, aimed at applications to

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		topology, geometry, groups and rings. Special emphasis on hom and tensor functors; projective, injective and flat modules; exact sequences; chain complexes; derived...
MATH237A	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials, Riemann-Roch theorem. Special topics may include:...
MATH237B	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials, Riemann-Roch theorem. Special topics may include:...
MATH237C	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials Riemann-Roch theorem. Special topics may include:...
MATH240A	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH240B	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH240C	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH241A	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of

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		gauge theories, geometric PDE, Morse theory and Flo...
MATH241B	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of gauge theories, geometric PDE, Morse theory and Flo...
MATH241C	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of gauge theories, geometric PDE, Morse theory and Flo...
MATH243A	Ordinary Differential Equations	Existence and stability of solutions, Floquet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling casc...
MATH243B	Ordinary Differential Equations	Existance and stability of solutions, Floq uet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling cas...
MATH243C	Ordinary Differential Equations	Existance and stability of solutions, Floquet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling casc...
MATH246A	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory an Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, elliptic regularity theory.

CODE	NAME	DESCRIPTION
MATH246B	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory and Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, elliptic regularity theory.
MATH246C	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory and Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, ellipitic regularity theory.
MATH260	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260A	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260AA	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260B	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260BB	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260C	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260CC	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260D	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260DD	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260E	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260EE	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260ES	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260F	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260G	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260GG	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260H	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional alalysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260HH	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260I	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260II	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260J	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260JJ	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260K	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260KK	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260L	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260LL	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260M	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260MM	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260N	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260NN	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260O	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260OO	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260P	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Q	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260QQ	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260R	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260S	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260SS	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260T	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260TT	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260U	Foundations in Mathematics	Topics in algebra, analysis, applied mathematics combinatorial mathematics,functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260UU	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260V	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260W	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260X	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Y	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Z	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH2A	Calculus with Algebra and Trigonometry	Math 3A with precalculus: A function approach integrating algebra, trigonometry, and differential calculus. Topics include: one-on-one and onto functions; inverse functions; properties and graphs of polynomial, rational, exponential, and logarithmic...
MATH2B	Calculus with Algebra and Trigonometry	Math 3B with precalculus: A continued review of relevant topics from precalculus (see Math 2A); integral calculus, including definite and indefinite integrals; techniques of integration, introduction to sequences and series; with applications in math...
MATH34A	Calculus for Social and Life Sciences	Introduction to differential and integral calculus with applications to modeling in the biological sciences.
MATH34B	Calculus for Social and Life Sciences	Continued study of differential and integral calculus with differential andintegral calculus with applications. Introduction to mathematical modeling with differential equations. Calculus of several variables including an introduction to partial deri...
MATH34LA	Calculus for the Life Sciences	Applications of differential and integral calculus to the life sciences. Includes mathematical foundations, biology examples, and computer programming to

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		simulate dynamic systems.
MATH34LB	Calculus for the Life Sciences	Applications of differential and integral calculus to the life sciences. Includes mathematical foundations, biology examples, and computer programming to simulate dynamic systems of more than one equation.
MATH3A	Calculus with Applications, First Course	Differential Calculus including analytic geometry, functions and limits, derivatives, techniques and applications of differentiation; introduction to integration; logarithmic and trigonometric functions.
MATH3B	Calculus with Applications, Second Course	Integral calculus including definite and indefinite integrals, techniques of integration; introduction to sequences and series; with applications in mathematics and physics.
MATH4A	Linear Algebra with Applications	Systems of linear equations, matrix algebra, determinants, vector spaces and subspaces, basis and dimension, linear transformations, eigenvalues and eigenvectors, diagonalization, and orthogonality.
MATH4AI	Inquiry Based Linear Algebra	Honors version of Mathematics 4A. Mathematical inquiry course is developed through problem solving and discovery.
MATH4B	Differential Equations	First and second order differential equations, separation of variables, linear differential equations, systems of first order equations, nonlinear differential equations and stability.
MATH4BI	Inquiry Based Differential Equations	Honors version of Mathematics 4B. Mathematical inquiry course is developed through problem solving and discovery.
MATH500	Teaching Assistant Practicum	Supervised teaching of undergraduate mathematics courses.
MATH501	Teaching Assistant Training	Consideration of ideas about the process of learning mathematics and discussion of approaches to teaching.

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MATH502	Teaching Associate Practicum	Supervised teaching of undergraduate courses.
MATH510	Reading for Area Examinations	Reading for area examinations.
MATH596	Directed Reading and Research	Directed reading and research.
MATH596AA	Directed Reading and Research	Directed reading and research.
MATH598	Master's Thesis Research and Preparation	Master's thesis research and preparation.
MATH598AA	Master's Thesis Research and Preparation	Master's thesis research and preparation.
MATH599	Dissertation Preparation	Dissertation preparation.
MATH6A	Vector Calculus with Applications, First Course	Calculus of functions of several variables, vector valued functions of one variable, derivative and integrals of vector functions, double and triple integrals, properties and applications of integrals, change of variables.
MATH6B	Vector calculus with Applications, Second Course	Scalar and vector fields, integration along paths, integration over surfaces and solid regions, integral theorems of vector calculus, Fourier series, partial differential equations.
MATH7H	Honors Seminar-Calculus	Emphasizing fundamental concepts and applications. Intended for highly motivated and well prepared students.
MATH8	Transition to Higher Mathematics	Introduction to the elements of propositional logic, techniques of mathematical proof, and fundamental mathematical structures, including sets, functions, relations, and other topics as time permits. Mastery of this material is essential for students...

CODE	NAME	DESCRIPTION
MATH94	Group Studies in Mathematics	Lectures and discussions on special topics. Designed for transfer students only.

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